

Tim McCloskey

XR / VR / Graphics Engineer · C · OpenGL · OpenXR · Meta Quest 3

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SUMMARY

Graphics engineer with deep roots in C and OpenGL, spanning scientific 3D visualization in the 1990s through a current VR project built entirely in raw C, OpenGL, and OpenXR. Designed and built CastVR — a FreeCAD-to-VR mechanical assembly visualizer on Meta Quest 3 — from the scene graph and PBR shader through to controller interaction and image-based lighting. Supplementary experience in Python, web development, and data pipeline work. Immediately available in Denver, CO.

CORE SKILLS

Graphics: C, OpenGL 4.x, GLSL, Cook-Torrance PBR, Procedural IBL, OpenXR, Assimp, cglm

VR: Meta Quest 3, OpenXR runtime, stereoscopic rendering, controller ray picking, 6DOF interaction

Tools: FreeCAD, Blender, Git, Make, GDB, Linux, Windows

Additional: Python, PHP, Java, JavaScript, PostgreSQL, MySQL, Docker, Agile

FEATURED PROJECT

CastVR: Universal CAD-to-VR Visualizer · 2024–2025 · github.com/TimVMcCloskey · youtu.be/nq0Fe0w1MXU

- ▶ Built entirely in raw C with OpenGL and OpenXR — no game engine, no VR framework
- ▶ Imports FreeCAD mechanical assemblies via Assimp; JSON-driven scene graph for multi-part assemblies
- ▶ Full PBR pipeline: Cook-Torrance BRDF, procedural IBL (irradiance + prefilter cubemaps + BRDF LUT), cached to disk
- ▶ Working mechanical linkage: Quest 3 controller drives pipe vise jaws through kinematic handle mechanism
- ▶ Explode / assemble, controller ray picking in model space, VR recenter with yaw/pitch orbit correction
- ▶ Stereoscopic headset render + roll-stripped PC mirror window with fixed distance camera for demo recording

PROFESSIONAL EXPERIENCE

Software Engineer · Voyant Communications · Englewood, CO | Feb 2019 – Mar 2020

- ▶ Migrated legacy PHP backend to Python 3; ported VM applications to Docker microservices
- ▶ Wrote a Python email parser consuming RabbitMQ messages to extract and re-queue embedded faxes
- ▶ Configured nginx proxy server for URL rewriting across production services

Software Developer · Great-West Financial (via Ascent Services) · Denver, CO | Sep 2018 – Feb 2019

- ▶ Built Python scripts to extract and analyze data from the corporate Jira database
- ▶ Ported and debugged MySQL query scripts from Windows to Linux server

Software Developer · NSF Center for Novel High Voltage Materials · Denver, CO | Sep 2017 – Aug 2018

- ▶ Wrote MATLAB modules computing fragment size from images of exploded electrical bushings
- ▶ Developed a Java application calculating positional deviation of glass fragments under ballistic impact
- ▶ Built a Python program to transform and triangulate experimental data into finite element format

Software Developer · EduCyber · Wheat Ridge, CO | Nov 2016 – Sep 2017

- ▶ Added features to custom WordPress plugins; tracked down and fixed ecommerce database bugs causing tax accounting errors on live sites

Graphics Engineer · Bedford Institute of Oceanography / Atlantic Geoscience Centre · Dartmouth, NS | 1992 – 1998

- ▶ Wrote scientific visualization software in C and OpenGL mapping geophysical data onto 3D ocean floor models
- ▶ Built a general-purpose map projection transformation application for oceanographic datasets
- ▶ Contributed to seismic horizon detection image processing software
- ▶ Wrote a real-time data logger integrating GPS navigation with research ship instrument arrays

EDUCATION & TRAINING

Associate Degree, Drafting · Siebel School of Drafting, Denver, CO

Udacity Full Stack Web Developer Nanodegree · 2018